Sourabh Deshpande web: sopdeshpande.github.io email: deshpasb@mail.uc.edu

## RESEARCH INTERESTS

Building and Deploying AI Solutions for Industrial Automation

- Vision-Language Models and Generative AI for human motion modeling in production and logistics environment
- Computer Vision for predicting product quality; Cloud-Edge / IoT solutions for process automation
- Operational and Human digital twins for safe human-object interactions in the physical world
- AI-driven Augmented / Virtual / Extended Reality for interactive learning
- Synthetic data generation for deep learning applications

## EDUCATION

**Ph.D.**, University of Cincinnati, OH, USA. GPA 4.0 (Advisor: Prof. Sam Anand) 2018 – Present
**Dissertation:** Machine Learning and IIoT-assisted Computer Vision Frameworks for Smart and Safe Manufacturing

**M.S.**, Mechanical Engineering, Michigan Technological University, MI, USA. 2015 – 2017

**B.S.**, Mechanical Engineering, University of Pune, India. 2010 – 2014

## INDUSTRY RESEARCH PROJECTS

**Siemens Technology, NJ, USA**

Modeling Human Motion for Improving Worker Safety in Manufacturing and Logistics 2022 - Present

- Trained an in-domain multi-task GPT model for automatically:
  - generating a 3D motion illustrating how to perform a new task ergonomically,
  - providing textual feedback on whether the worker's current motion could lead to injury,
  - predicting the worker's projected path to prevent any object collisions
- Devised LLM-assisted toolbox to label custom human motions of factory workers with ergonomic descriptions
- Project to be deployed at selective industrial sites through Ohio Bureau of Workers' Compensation state agency

Automated Quality Control using AI and Computer Vision (CV)

- Led the group to develop a closed-loop CV system for automated inspection of sealant application for space-grade solar panel
- Performed object localization and image segmentation of the sealant deposited on plates in real-time, and trained a Bayesian network to provide corrective action recommendations.
- Deployed at Boeing facility, Charleston, SC as part of DMDII/MxD grant

**Siemens, P&G, Kinetic Vision, Worthington Steel** (Selective Industry 4.0/5.0 Institute members at UC) 2023 - 2024

Enhancing Digital Twins and IoT for Legacy and Smart Factory Machines with Large Language Model (LLM) Feedback

- Co-developed LLM-based digital twin for guiding new users to control factory machines (on-site / remotely) through voice commands
- Proposed a RAG-based multi-agent framework to process IoT data for visualizing plots, generating documentation for machine operations, conducting fault diagnosis, and suggesting potential fixes (Latency < 1.5 seconds)

Improving Worker Manufacturing Training with Augmented Reality and Deep Learning

- Developed an AR-based training framework that simplifies complex component identification for shop floor workers
- Built a novel synthetic data generation pipeline, based on CAD models using physics-based rendering methods, and fine-tuned a deep learning model (U-net) for pixel-level component identification and performing assembly

**Volvo Trucks** 2021

Real-Time Visualization of Machine Sensors for Factory Maintenance using Cloud and AR Application

- Co-developed an AR methodology to locate sensor positions (spatial anchors) on machines and visualize data in real-time to aid workers
- Reduced machine troubleshooting and downtime by 30%, translating to several million dollars per year in savings

**Raytheon Technologies** 2020

CAD-Based Computational Tools for Metal Powder Additive Manufacturing

- Developed computational geometry algorithms for feature recognition in 3D geometries
- Designed frontend application GUI within Siemens NX

**Tata Motors Ltd.** 2015

Engineering Intern, Research and Development

- Designed rapid prototype and sheet metal components in CAD for 3D printing and press bending operations

## RELEVANT RESEARCH PROJECTS

### Industrial IoT for Smart Factory

- Created an edge and cloud-based AR framework to display and control real-time legacy machine data on an iPad and provide corrective actions using a Bayesian Network; awarded $40K grant from UC Industry 4.0/5.0 Institute to build a Digital Twin

### Prediction of Selective Laser Melting (3D Printing) Part Quality using Hybrid Bayesian Network

- Devised a Bayesian network to model the correlation between process parameters and part quality attributes of SLM machines

### Deep Learning Segmentation Methods for Defect Detection in Additive Manufacturing

- Published a survey at Int. Journal of Advanced Manufacturing Technology on CV methods reported for 3D printing using YOLO, SSD, Faster R-CNN, U-Net, Mask R-CNN, GAN, and Vision Transformers and implementing them for closed-loop monitoring

## PROGRAMMING AND SOFTWARE SKILLS

Programming languages: Python, C#, C++, MATLAB, JavaScript, HTML, SQLite
Commercial software: NX, Siemens Jack/Process Simulate, Ansys Additive, Autodesk Netfabb, Cortex
Open-source libraries: OpenCV, PyTorch, Tensorflow, Blender, ONNX, GitHub, HuggingFace

## SELECTIVE PUBLICATIONS

IIoT-Enabled Digital Twin or Legacy and Smart Factory Machines with LLM Integration 2025
Anuj Gautam, Manish Aryal, **Sourabh Deshpande**, Shailesh Padalkar, Ming Tang, Sam Anand, et al.
*SME North American Manufacturing Research Conference (Accepted)*

Deep Learning-Based Recognition of Manufacturing Components using AR for Worker Training of Assembly Tasks 2024
**Sourabh Deshpande**, Manish Aryal, Sam Anand
*ASME Manufacturing Science and Engineering Conference Proceedings,* https://doi.org/10.1115/MSEC2024-125279

ImVR: Immersive Design Exploration and Process Integration for Additive Manufacturing of Complex Organic Geometries 2024
Manish Aryal, **Sourabh Deshpande**, Jillian Aurisano, Sam Anand
*ASME Manufacturing Science and Engineering Conference Proceedings,* https://doi.org/10.1115/MSEC2024-121253

Smart Monitoring and Automated Real-Time Visual Inspection of a Sealant Applications 2023
**Sourabh Deshpande**, Aditi Roy, Joshua Johnson, Ethan Fitz, Manish Kumar, Sam Anand
*SME Manufacturing Letters,* https://doi.org/10.1016/j.mfglet.2023.08.115

IIoT Based Framework for Data Communication And Prediction Using Augmented Reality For Legacy Machine Artifacts 2023
**Sourabh Deshpande**, Shailesh Padalkar, Sam Anand
*SME Manufacturing Letters,* https://doi.org/10.1016/j.mfglet.2023.08.058

IIot Based Augmented Reality for Factory Data Collection and Visualization 2021
Jonathan Rosales, **Sourabh Deshpande**, Sam Anand
*Procedia Manufacturing,* https://doi.org/10.1016/j.promfg.2021.06.062

Prediction of Selective Laser Melting Part Quality Using Hybrid Bayesian Network 2020
Nathan Hertlein, **Sourabh Deshpande**, Vysakh Venugopal, Manish Kumar, Sam Anand
*Additive Manufacturing,* https://doi.org/10.1016/j.addma.2020.101089

## RELEVANT COURSEWORK

- Intelligent Systems (AI), Introduction to Industrial AI, Statistical Methods
- Math Methods for Decision Engineering, Advanced Quality Engineering, Computational Methods in Additive Manufacturing

## ACADEMIC MENTORSHIP

- Manish Aryal (M.S. Thesis) – ImVR: Enabling Immersive Exploration Environment for Complex Organic Geometries, 2023
- Anuj Gautam (M.S. Thesis) – IIoT-enabled Digital Twin with LLM Feedback for Manufacturing Applications, 2024

## ACCOMPLISHMENTS AND LEADERSHIP EXPERIENCE

- University of Cincinnati Office of Research Digital Futures Fellowship, 2024
- Invention Disclosure (UC), along with PI Prof. Anand and Co-PI Prof. Kumar - Object Localization MxD, 2022
- NSF student travel award for attending and recognition by SME and ASME for co-organizing NAMRC and MSEC, 2021
- Graduate teaching assistant, MECH 6071, Advanced Design for Manufacturing, 2019
- Gold and bronze medal at national level cycling tournament, School Games Federation of India, 2010
- Best All-Around and Best Character award, Tata Motors Employees Education Trust, India, 2008